

E-Commerce Sales Analysis

Tool Used: Microsoft Excel

Skills: Data Cleaning, Pivot Tables, Pivot Charts,
Dashboard Design, Business Analysis

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Data Analyst intern
Dataset



content

Project Overview



Geographic Analysis



Data Cleaning



Product Performance



KPI Metrics



Filters Explanation



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Key Insights



Category Analysis



Recommendations



Project Overview

Objective

The objective of this project is to analyze e-commerce sales performance and provide insights into sales trends, profitability, customer segments, and regional performance.

Business Problem

E-commerce companies generate large amounts of sales data. Without proper visualization, it becomes difficult to understand:

- Which product categories generate the most revenue
- Which regions perform best
- Which subcategories drive sales
- Year-over-Year growth trends
- Profitability performance

This dashboard solves this problem by presenting interactive insights in a single view.

**a deep understanding of
consumer behavior**



Dataset Information

The dataset includes the following fields:

- Order Date
- Sales
- Profit
- Quantity
- Category
- Sub-Category
- Region
- State
- Segment

Data Preparation

Data Cleaning Steps

- Removed duplicate records
- Standardized date format
- Checked for missing values
- Created calculated columns
- Structured dataset for pivot tables



Data Model

Key Metrics Created:

- Total Sales
- Total Profit
- Total Quantity
- Number of Orders
- Profit Margin %
- YoY Growth %

Key Performance Indicators

KPI	Value
Total Sales	\$ 2297200.86
Total Profit	\$ 286397.0217
Quantity Sold	# 37873
Number of Orders	# 9994
Profit Margin	12.47%

Each KPI also includes Year-over-Year (YoY) growth which indicates performance improvement compared to the previous year.

- Sales increased by 20.62%
- Profit increased by 14.41%
- Quantity sold increased by 27.45%
- Orders increased by 28.64%
- Profit margin slightly declined by 5.15%



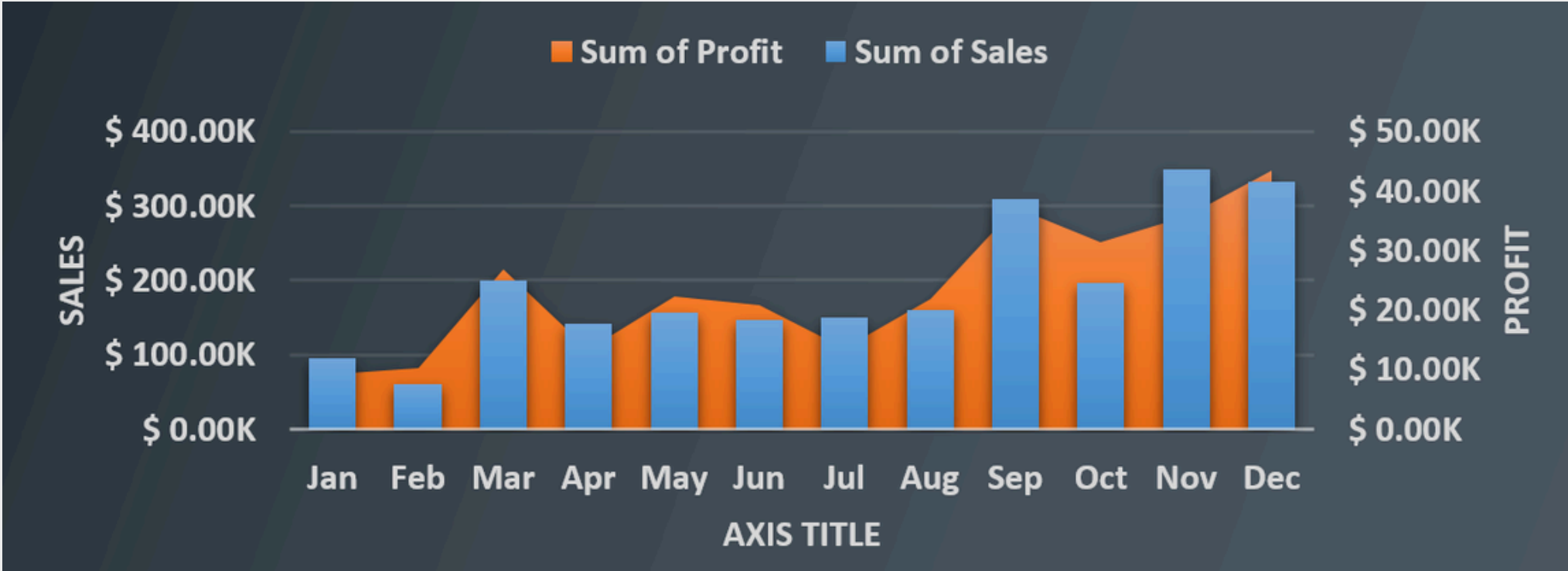
Sales and Profit Trend Analysis

Description

This chart compares monthly sales and profit performance across the year. It helps identify trends, seasonal patterns, and periods of strong or weak performance.

Two metrics are displayed

- Sales (Revenue generated)
- Profit (Actual earnings after cost)



Insight

February shows high profit performance.

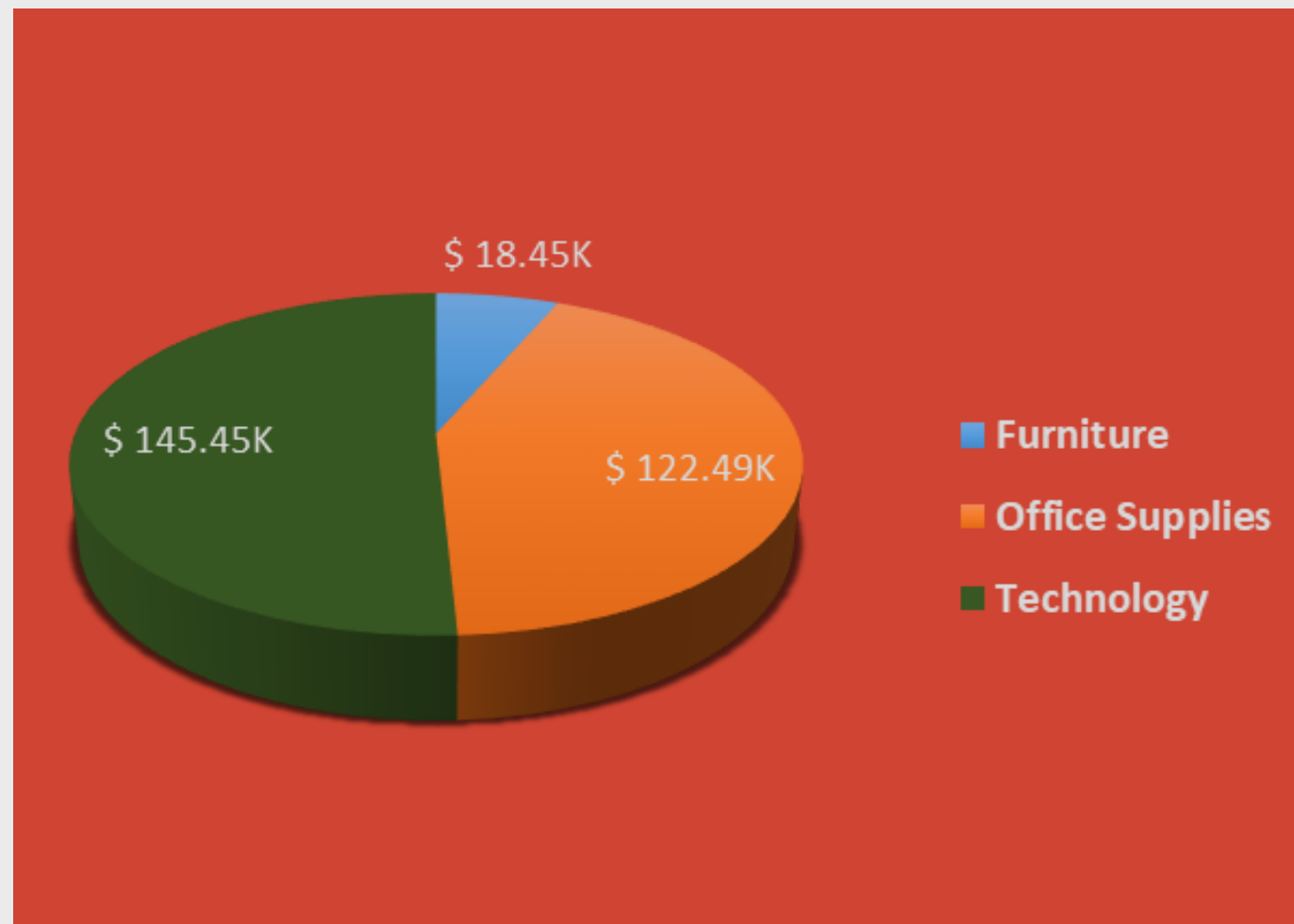
November and December show strong sales growth, indicating seasonal demand.

Some months have high sales but moderate profit, indicating higher costs or discounts.

Business Value

This chart helps businesses understand seasonal demand patterns and profitability trends.

Category Performance Analysis



Insight

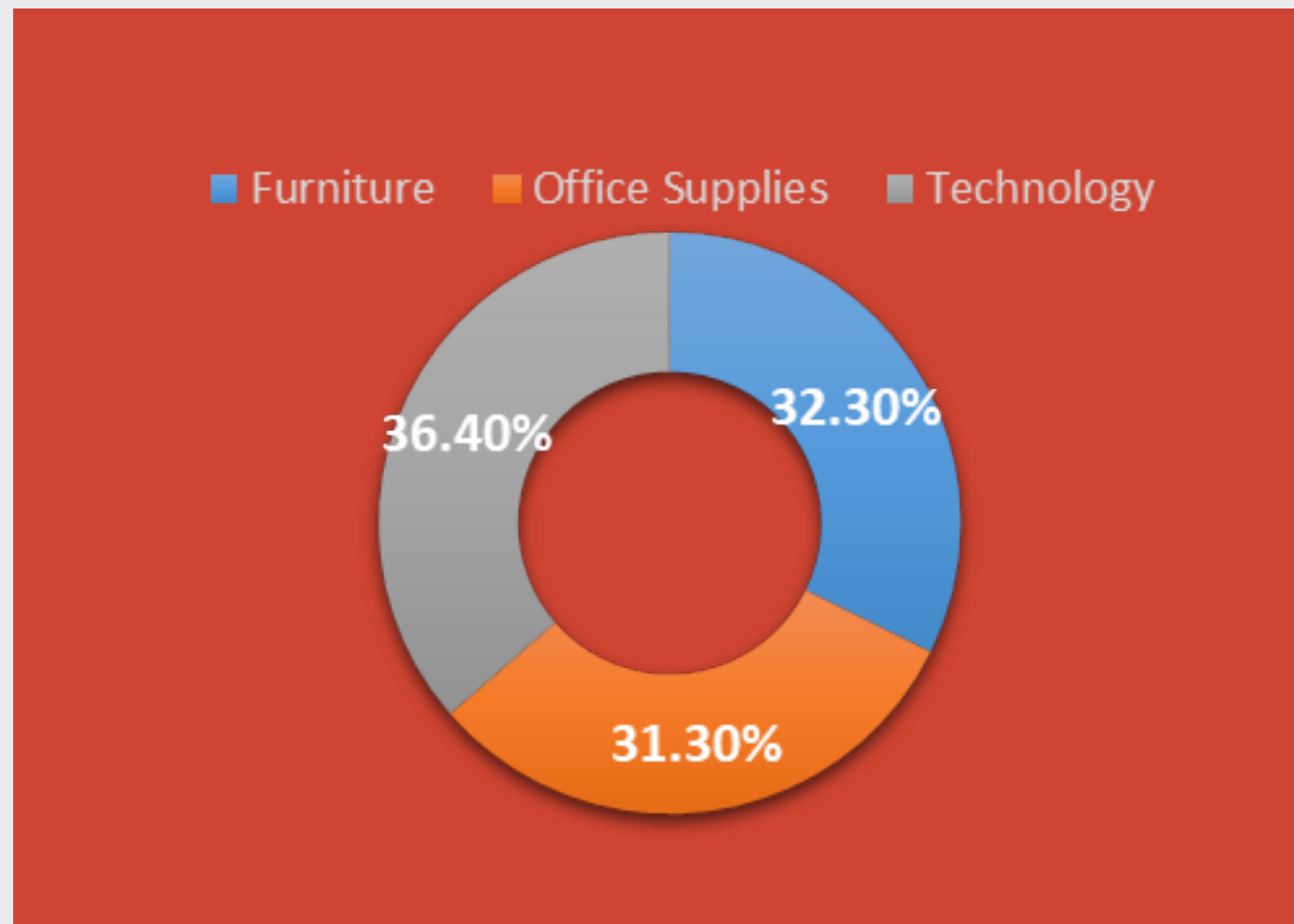
Technology generates the **highest profit**.
Office Supplies contributes a **moderate** share.
Furniture generates the **lowest profit** among categories

Business Value

This chart helps businesses identify **which product category contributes the most profit** and where improvement is needed.

Category Wise Sales Shares

The donut chart represents the **percentage contribution of each category to total sales.**



Insight

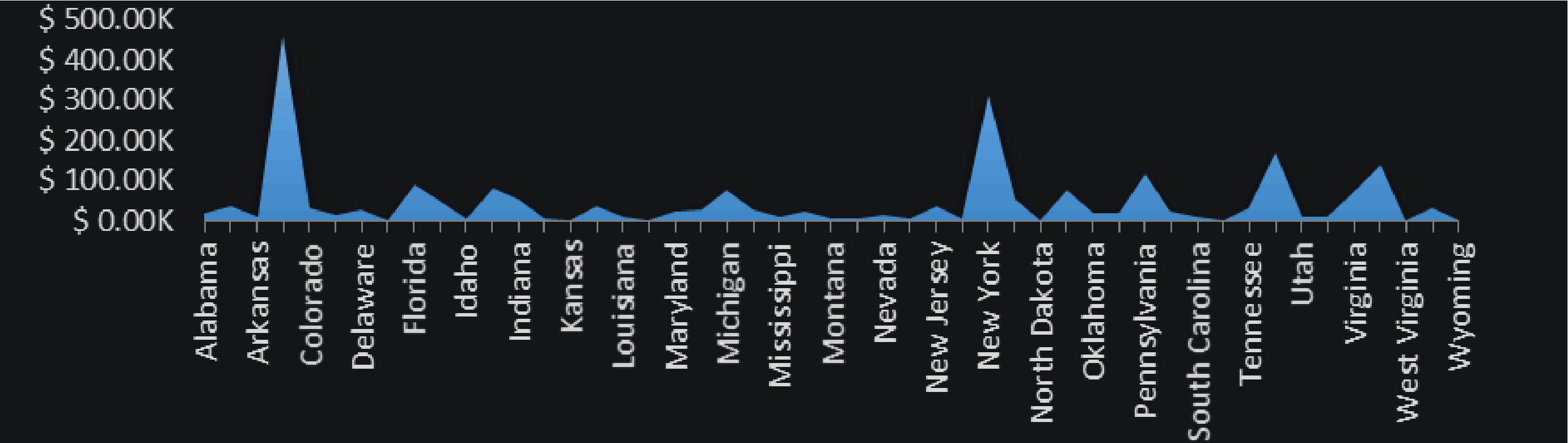
Technology contributes the **largest share of total sales**, indicating strong customer demand.

Business Value

Helps management understand **which product category drives the most revenue.**

Geographic Sales Analysis

Shows which states contribute most revenue.



Insight

Virginia shows the **highest sales**
Louisiana has the **lowest sales**

Business Value

Helps companies identify **high-performing and low-performing regions** for better marketing and expansion strategies.

Product Performance

This chart shows the **top five product subcategories based on sales revenue.**

Insight

Machines generate the **highest sales revenue.**
Tables and Phones also contribute significantly.
Accessories have the **lowest sales among the top five**



Business Value

Helps companies identify **best-selling products** and prioritize inventory and marketing strategies.



Interactive Filters

Dashboard includes slicers for

Year
Region
Segment



Purpose

These filters allow users to dynamically analyze:

- Sales by different years
- Performance across regions
- Customer segments

Segments include:

- Consumer
- Corporate
- Home Office

Category Performance Analysis

Key Business Insights

Major Findings

1. Technology category generates the highest profit.
2. Machines are the top performing subcategory.
3. Sales peak in November and December.
4. Virginia shows strong regional sales.
5. Profit margin needs improvement despite sales growth.

Recommendations

Based on analysis:

1. Increase focus on Technology products.
2. Improve profitability in Furniture category.
3. Expand marketing in high-performing states.
4. Promote top-selling subcategories like Machines.
5. Improve pricing strategy to increase profit margin.

thank you

DASHBOARD

ECOMMERCE SALES ANALYSIS DASHBOARD

SALES

\$2,297,200.86



YOY GROWTH ▲ 20.62%

PROFIT

\$286,397.02



YOY GROWTH ▲ 14.41%

QUANTITY

37873



YOY GROWTH ▲ 27.45%

NO OF ORDERS

9994



YOY GROWTH ▲ 28.64%

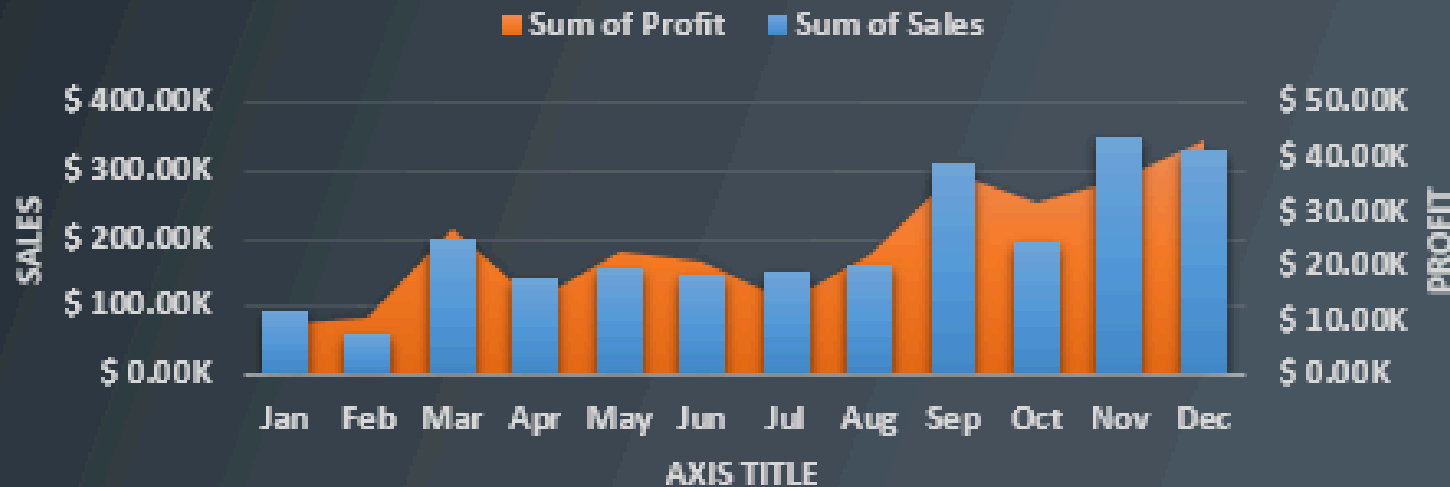
PROFIT MARGIN

12.47%

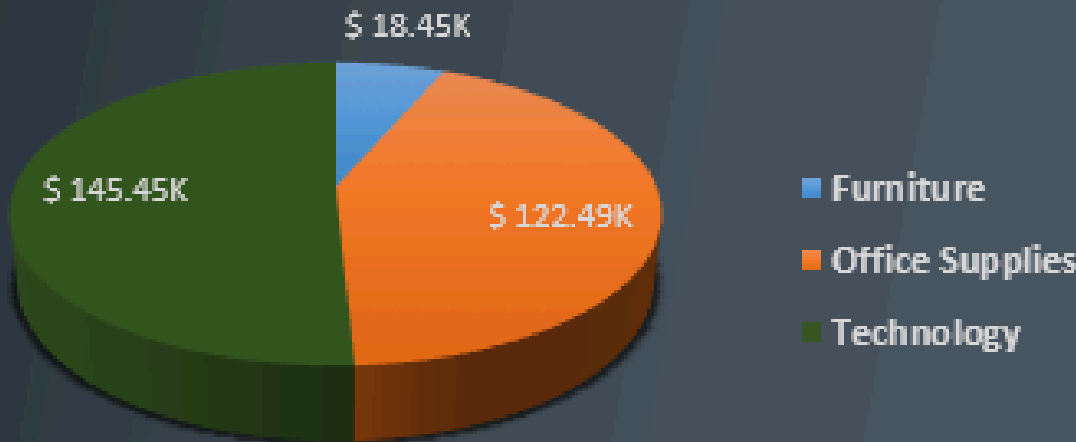


YOY GROWTH ▼ 5.15%

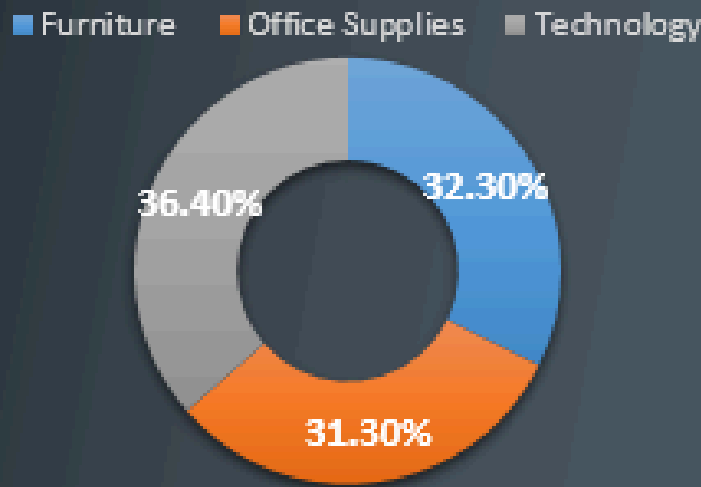
SALES AND PROFIT ANALYSIS



CATEGORY WISE PROFIT



CATEGORY WISE SALES SHARE %



SALES BY STATE



TOP 5 SUBCATEGORIES BY SALES



FILTERS

Year	Region	Segment
2011	Central	Consumer
2012	East	Corporate
2013	South	Home Office
2014	West	